

DAEMONCORE

FieldOps Capability and Technology Guide

Professional assessment operations for authorized targets

RELEASE 6.5.1 WINDOWS AND LINUX SEPTEMBER 2026

A complete buyer and operator reference to FieldOps diagnostics, managed tools, resilience testing, evidence controls, findings, reporting, supported technologies, deployment requirements and current boundaries.

CONTROLLED PROFESSIONAL USE. A software license does not authorize a target. Operate only under valid written authorization, exact scope, an active testing window and an agreed stop procedure.

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DaemonCore FieldOps is an authorization-bound assessment workspace for professional operators. It brings scope, execution, evidence, findings, resilience testing, and reporting into one local-first desktop record. This guide describes the capabilities shipped in release 6.5.1, the technologies FieldOps works with, and the operational limits buyers and operators should understand.

FieldOps does not grant authority over any system. A license unlocks the software. A valid engagement, signed Rules of Engagement, target-owner approval, and applicable law govern the work.

Capability status key

STATUS	MEANING
Native	FieldOps performs the operation directly and records the result.
Managed integration	FieldOps launches a supported external engine with fixed arguments, streams output, and seals successful results.
Evidence bridge	FieldOps detects the external tool, exports signed scope, and imports supported structured results; the operator runs the tool separately.
Academy range	Available as a contained learning or practice workflow, not as a FieldOps production runner.
Not currently provided	Not executed or orchestrated by FieldOps 6.5.1.

What FieldOps is built to answer

FieldOps is designed around questions an assessor must be able to defend:

- What exact system was authorized, resolved, and reached?
- Which TCP services were reachable inside the declared port set?
- What did the service identify itself as, and how confident is that identification?
- What HTTP, TLS, and certificate posture was observed from this vantage point?
- What changed since the prior surface baseline?
- How did the service behave under an owner-approved workload?
- Which evidence supports the finding, impact, remediation, and retest decision?
- Who operated the assessment, who approved it, and when did every action occur?

Assessment lifecycle coverage

PHASE	FIELDOPS CAPABILITY	OUTPUT
Prepare	Named operator enrollment, device key, engagement definition, approver and authorization reference	Operator identity and signed operation permit

PHASE	FIELDOPS CAPABILITY	OUTPUT
Constrain	Exact targets, TCP ports, network boundary, time window, policy and execution capacity	Enforced authorization boundary
Discover	DNS, TCP reachability, allowed-port survey and service profiling	Digest-sealed captures
Characterize	HTTP response and posture, TLS identity, web mapping, deep service inventory	Asset and service evidence
Compare	Complete surface baseline and prior-versus-current drift analysis	Change intelligence
Validate	Managed Nmap or imported specialist-tool evidence	Tool-attributed evidence record
Exercise	Managed k6 and bounded Chaos Engine resilience profiles	Workload telemetry and signed receipt
Assess	Findings, severity, impact, remediation, disposition and retest	Reviewed finding register
Report	Printable report and machine-readable case export	Client report and JSON evidence bundle
Close	Scope closure with evidence retained and execution blocked	Durable closed engagement

Authorization and operator identity

Named operator enrollment

Status: Native

FieldOps records the operator's name, organization and professional email, then creates an Ed25519 key pair bound to the local installation. The public fingerprint is included in signed permits and operation receipts. The private key is protected through operating-system credential storage when supported.

Operator attribution is tamper-evident, not independent identity verification. A typed identity must still be checked through the organization's normal personnel and engagement process.

Signed operation permits

Status: Native

Each new engagement can bind:

- client and engagement name;
- named approving authority and professional email;
- authorization reference;
- internal or external network mode;
- exact targets and TCP ports;

- validity start and end times;
- Observe, Validate or Stress policy level;
- Guarded or Professional execution profile;
- permitted operations and execution capacity.

FieldOps verifies the permit signature, active time window, operation allowance, and agreement between the current engagement record and the signed permit before execution.

Destination resolution and pinning

Status: Native

FieldOps resolves authorized hostnames and pins an operation to a resolved address. External engagements reject loopback, private, link-local, multicast, reserved, documentation and mixed-boundary results. Internal engagements accept private address space and reject public destinations. HTTP operations connect to the pinned address while retaining the authorized hostname for Host and TLS server-name checks.

Native network diagnostics

CAPABILITY	TECHNOLOGY	WHAT FIELDOPS RECORDS
DNS resolution	Operating-system DNS resolver	Authorized hostname and resolved IPv4 or IPv6 addresses
DNS profile	DNS A, AAAA, CNAME, MX, NS and TXT queries where available	Record sets, aliases and mail or name-service context
TCP reachability	TCP connect	Connection success or failure and observed latency
Allowed-port survey	Concurrent TCP connect across the signed port list	Open, closed or rejected, and filtered or unresponsive observations
HTTP response	Pinned HTTP or HTTPS HEAD request	Status, status text and bounded response headers
HTTP security posture	Header and transport analysis	Evidence-based posture signals and observations
TLS identity	Node TLS client	Protocol, cipher, ALPN, trust result, subject, SAN, issuer, validity and SHA-256 fingerprint
Service profile	Banner, HTTP and TLS correlation	Likely service, evidence signal and confidence
Bounded web map	Approved-path HTTP inspection	Reachable web-path observations inside the declared endpoint
Basic resilience baseline	Ten spaced HEAD samples	Success count, response durations and average duration

FieldOps does not treat a port number or banner as proof of a vulnerability. Results are observations that an operator must interpret and, where authorized, validate.

Asset intelligence and surface comparison

Complete target baseline

Status: Native

The complete baseline combines address resolution, allowed-port state, service observations, HTTP posture and available TLS evidence for one authorized target. A later baseline can be compared with the prior sealed capture.

FieldOps change intelligence can highlight:

- added or removed addresses;
- opened, closed or state-changed declared ports;
- changed HTTP posture signals;
- changed TLS certificate identity or validity details;
- material differences in the observed web surface.

The Evidence workspace groups captures by authorized asset, displays the latest observed services, and exposes material drift for review.

Deep service inventory

Status: Native with managed-engine support

Deep inventory examines the signed port set and builds structured service records. When managed Nmap is selected, FieldOps uses service and version detection to normalize protocol, state, reason, product, version, tunnel, CPE and confidence information.

Managed Nmap execution

Status: Managed integration

FieldOps can use a locally installed Nmap executable or its pinned Docker adapter. The managed runner:

- accepts one already resolved and authorized IP address;
- accepts only the engagement's normalized TCP port list;
- invokes the process directly without a command shell;
- uses TCP connect scanning with host discovery and DNS disabled;
- requests service and version detection with bounded retries;
- streams process output into the War Room;
- supports operator cancellation and execution timeout;
- parses Nmap XML into structured service evidence;
- records the engine and version;
- seals a successful result into the engagement evidence chain.

The Docker fallback uses `instrumentisto/nmap:7.98-r2` and requires a running Docker engine. Docker may retrieve the image on first use.

FieldOps 6.5.1 does not expose arbitrary Nmap flags or shell text through the managed runner.

Tool and technology matrix

TECHNOLOGY	FIELDOPS RELATIONSHIP	PRIMARY USE
Nmap	Managed integration; native executable or Docker adapter	TCP service and version inventory
Docker Engine	Native runtime integration	Nmap adapter and contained Academy ranges
Grafana k6	Managed integration	Verified HTTP workload and resilience execution
ProjectDiscovery Nuclei	Evidence bridge	Template-driven validation evidence
Locust	Evidence bridge	Customer-managed workload evidence
Wireshark and TShark	Evidence bridge	Packet and protocol evidence
Hashcat	Evidence bridge	Offline credential-audit evidence
JSON	Native evidence intake and case export	Structured external results and engagement records
SARIF	Native evidence intake	Static and security-analysis result exchange
HTML	Native report export	Printable client-facing engagement report
SHA-256	Native integrity control	Capture, source-file and audit-chain digests
Ed25519	Native signing control	Operator permits and operation receipts
OS credential storage	Native platform integration	Protection of licensing and FieldOps signing secrets

Availability detection reports the installed tool version where possible. Detection does not mean FieldOps manages every capability of that tool.

Signed scope export and evidence intake

Execution manifests

Status: Native

For supported tools, FieldOps exports a signed execution manifest containing the engagement, tool, target boundary, ports, policy, permit reference and applicable workload information. This gives an operator a portable scope artifact for a separately operated specialist tool.

Evidence bridges

Status: Native intake for externally executed tools

FieldOps can import JSON or SARIF evidence attributed to Nuclei, Locust, TShark, Hashcat and other cataloged tools. The intake workflow:

- never executes imported content;
- validates the selected format and structure;
- records the original filename and SHA-256 source digest;
- binds the result to one exact authorized target;
- records the source tool and version context when supplied;
- seals the imported document as a reviewable capture.

Raw packet captures, password-cracking sessions and arbitrary tool output are not executed by the evidence bridge. Preserve source artifacts using the client's approved evidence-storage process.

Assessment campaigns

Status: Native

Campaign Control coordinates repeatable approved checks across selected targets in one engagement. Shipped profiles include:

PROFILE	COORDINATED OPERATIONS
Service Inventory	Deep service inventory
Surface Verification	DNS profile and complete surface baseline
Complete Assessment	DNS profile, deep inventory and complete surface baseline

Campaigns maintain a durable task ledger and progress summary. Operators can request pause, resume interrupted or failed work, and cancel safely. An active module is allowed to finish before pause or cancellation completes, preventing an abrupt partial write from being represented as complete evidence.

Managed load and resilience testing

Target-hosted capacity verification

Status: Native control

Before managed load execution, the exact target must publish a challenge-bound JSON capacity grant at a DaemonCore well-known path. FieldOps verifies the target, authorization reference, challenge, transport, port, source address, expiration, maximum requests per second, maximum concurrency and maximum duration.

External grants require verified TLS. The target must continue resolving to the address that issued the grant when execution begins.

Managed Grafana k6

Status: Managed integration

FieldOps generates a fixed, shell-free HTTP GET workload and launches a locally installed k6 process. Supported profiles are:

PROFILE	INTENDED QUESTION
Ramp	How does behavior change as arrival rate rises and falls?
Spike	How does the service absorb and recover from a short abrupt increase?
Soak	Does sustained approved demand reveal time-dependent degradation?
Breakpoint	Where does the first unacceptable behavior appear within the grant?
Recovery	Does the service return to the agreed state after sustained demand?

The plan binds target, resolved address, port, path, transport, requested RPS, duration, concurrency, p95 threshold, error threshold and capacity-grant digest. Redirects are disabled and response bodies are discarded.

Live and final telemetry

FieldOps records or displays:

- total HTTP requests and achieved RPS;
- p50, p90, p95 and p99 request duration in the final summary;
- error rate;
- dropped iterations;
- maximum virtual users;
- passed and failed checks;
- engine identity and version;
- process output, lifecycle state, stop reason and outcome.

The War Room displays live request count, achieved RPS, p95 latency, error rate and dropped iterations as data becomes available. Target-side CPU, memory, queues, pools and dependency saturation require the customer's monitoring platform; FieldOps does not invent target telemetry it cannot observe.

Stop controls

Status: Native

Managed load provides an immediate local process stop. FieldOps also polls a challenge-bound target-side stop endpoint; the system owner can stop the active run or all challenge-bound runs. A completed or stopped workload is sealed with its signed receipt, plan, engine information, metrics and outcome.

Chaos Engine

Status: Native bounded resilience sampler

The Chaos Engine runs short HTTP HEAD-based resilience experiments against one authorized endpoint. It supports baseline, ramp, spike, soak and recovery profiles, captures latency and error behavior, calculates a resilience score, applies automatic p95 and error-rate abort criteria, and retains an experiment ledger.

Chaos Engine is intentionally distinct from managed k6. It is a bounded workstation-side sampler for rapid validation. Managed k6 is the capability for owner-verified workload execution under a target-hosted capacity grant.

Chaos Engine 6.5.1 is not a host-fault, process-kill, packet-loss, infrastructure-failure or distributed denial-of-service framework.

Evidence integrity and auditability

Capture integrity

Status: Native

Every FieldOps capture includes a SHA-256 digest over its stored content. Snapshot verification detects a changed capture. Imported evidence also retains the source-file digest.

Activity ledger

Status: Native

Engagement activity is written to a hash-linked audit chain. Entries record operation, status, summary, evidence reference and timestamp. Where operator signing is configured, operation receipts are signed with the device-bound Ed25519 key.

Integrity semantics

FieldOps provides tamper evidence and operator attribution. It does not claim that a local record is independently witnessed, legally admissible, immutable against an administrator, or proof that a typed operator identity and authorization claim are truthful. Export promptly and place important case material under the client's normal evidence-retention controls.

Findings and remediation verification

Status: Native

Operators can promote captures into findings and record:

- affected target and concise condition title;
- severity from informational through critical;
- observation and supporting evidence;
- credible impact and limitations;
- remediation guidance;
- open, accepted, remediated or closed disposition;
- evidence-backed retest history;
- fixed or still-present retest verdict.

FieldOps prevents a finding from pretending that unsupported tool output is proof. The quality of the conclusion still depends on the operator's technical judgment, authorization and evidence.

Reports and portable case records

Professional report

Status: Native

The printable HTML report includes engagement and authorization metadata, named operator and approving authority, integrity state, finding counts, campaign summary, service inventory, surface-change intelligence, finding details and retest counts.

Machine-readable case export

Status: Native

The JSON case bundle includes the engagement, operator identity, managed-tool jobs, load runs, campaigns, captures, findings, Chaos Engine runs, audit entries and integrity results. It is suitable for archival, independent review and downstream transformation.

Operating systems and storage

AREA	WINDOWS	LINUX
Desktop package	Installer and Microsoft Store package path	ApplImage and Debian package
FieldOps workspace	Supported	Supported
Docker-backed Academy ranges	Docker Desktop	Docker Engine or compatible runtime
Managed Nmap	Local executable or Docker adapter	Local executable or Docker adapter
Managed k6	Locally installed k6	Locally installed k6
Secret protection	Windows credential protection	Secret Service-compatible keyring such as GNOME Keyring or KWallet
Local persistence	Electron user-data directory	Electron user-data directory

Operator records, Academy progress, FieldOps case files and license state are designed to persist across application upgrades when the normal installer and package paths are used. Export important engagements before workstation migration, profile deletion or operating-system reinstallation.

Guarded and Professional execution profiles

CONTROL	GUARDED	PROFESSIONAL
Intended use	Focused validation	Experienced assessment teams
Target capacity	Up to 100 declared targets	Exact target list signed into the engagement
Port capacity	Up to 128 declared ports	Exact port list signed into the engagement

CONTROL	GUARDED	PROFESSIONAL
Port-survey concurrency	Four by default	Operator-selected within signed capacity, up to 64
Managed Nmap window	Six minutes by default	Configurable within signed capacity, up to 60 minutes
Authorization enforcement	Required	Required
Destination pinning and audit	Required	Required

Professional removes product-wide target and port cardinality ceilings; it does not remove authorization, boundary, time-window, process-stop, integrity or workstation-resource controls.

Hacker mindset in a professional workflow

FieldOps supports adversarial thinking without turning evidence into theater. The operator workflow is:

1. **Model the target.** Identify names, addresses, services, trust boundaries and dependencies inside scope.
2. **Question assumptions.** Compare declared architecture with observed DNS, service, HTTP and TLS evidence.
3. **Find the weak signal.** Look for unexpected exposure, identity drift, missing controls, inconsistent responses and fragile capacity.
4. **Prove the minimum.** Use the least invasive operation that establishes the condition and stop when the proof point is reached.
5. **Think in paths.** Connect exposure, prerequisite, control failure and consequence without overstating untested steps.
6. **Preserve uncertainty.** Record competing explanations and the evidence needed to resolve them.
7. **Retest the control.** Verify that remediation breaks the demonstrated path without breaking intended service behavior.

The mindset is offensive. The operating standard is controlled, attributable and reproducible.

What FieldOps 6.5.1 does not currently provide

For accurate procurement and operator expectations, the current release does not provide:

- arbitrary shell or unrestricted command execution through FieldOps;
- a managed exploitation-framework runner;
- payload generation, persistence or destructive post-exploitation automation;
- online password guessing or credential spraying;
- wireless, Bluetooth, radio-frequency or hardware-bus testing;
- active packet capture or packet injection through the TShark evidence bridge;
- managed Hashcat cracking sessions;
- managed Nuclei or Locust execution;
- cloud-provider API enumeration or cloud control-plane remediation;
- endpoint detection and response, malware analysis or host-forensics acquisition;
- vulnerability-feed subscription or automatic CVE enrichment;
- centrally orchestrated distributed load workers in release 6.5.1;

- independent approver identity verification or a hosted certificate authority;
- a replacement for client monitoring, ticketing, SIEM, evidence storage or legal review.

These boundaries distinguish shipped functionality from external-tool interoperability and future product direction.

Deployment prerequisites

Required

- DaemonCore Academy 6.5.1 on a supported Windows or Linux desktop;
- active FieldOps entitlement;
- named operator identity and protected local signing key;
- written authorization and exact engagement scope;
- network access to the authorized target;
- sufficient local storage for case data and exports.

Capability dependent

- Nmap installed locally, or Docker running for the managed Nmap adapter;
- Grafana k6 installed locally for managed load execution;
- target-owner capacity grant and stop endpoint for managed load;
- Nuclei, Locust, TShark or Hashcat for their respective external evidence workflows;
- customer monitoring for target saturation and recovery telemetry;
- Linux keyring service for protected FieldOps credentials on Linux.

Capability selection guide

OPERATOR QUESTION	RECOMMENDED STARTING CAPABILITY
Where does this name resolve?	DNS resolution or DNS profile
Is this exact declared service reachable?	TCP reachability
Which declared ports respond?	Allowed-port survey
What appears to be listening?	Service profile or managed Nmap
What does the web endpoint reveal without a body request?	HTTP response and posture
Does certificate identity match the intended service?	TLS identity
What changed since the last approved review?	Complete target baseline and change intelligence
Can I repeat the same checks across several targets?	Campaign Control
Can this service meet an approved traffic objective?	Verified managed k6
Can I obtain a quick low-volume resilience signal?	Chaos Engine
How do I preserve a specialist-tool result?	Signed scope export and structured evidence intake

OPERATOR QUESTION	RECOMMENDED STARTING CAPABILITY
How do I turn observations into client action?	Evidence promotion, findings and retest

Product positioning

FieldOps is best understood as an assessment operations system, not a single scanner. Its value is the continuity between authorization, technical action, evidence, decision and reporting. It can coordinate supported technologies while retaining the exact operator, target, scope and result context that stand-alone command output often loses.

The software is intentionally local-first. Customer-controlled systems perform the assessment work, customer monitoring supplies target-side truth, and exported records can move into the customer's existing governance and evidence systems.

Terminology and legal notice

DaemonCore product names such as FieldOps, War Room, Execution Fabric, Capacity Grant, Sealed Evidence and Chaos Engine are defined in the **DaemonCore Terminology Guide**. They do not replace the executed Rules of Engagement, Statement of Work, authorization, organizational policy or applicable law.

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